

KARTHIK A

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING ENGINEER

Bengaluru, India · +91-9741363230 · karthikanandofficial@gmail.com · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

SUMMARY

AI/ML Engineer with 6+ years designing, building and deploying production ML/LLM systems. Deep expertise in RAG, Agentic AI, and Prompt Engineering, with production systems in OCR/IDP, Face Auth, and Voice Auth. Strong foundation in Classical ML and Deep Learning. Comfortable owning systems end-to-end, from scalable microservice architecture to observability and security.

EXPERIENCE

StatusNeo	Bengaluru, India
Senior Consultant	June 2025 – Present
- Architected and led the bank's first production Python AI platform: document classification/extraction microservices for 800+ users pan-India at 97% accuracy. - Co-designed an agentic banking platform orchestrating workflows through MCP-compatible tool layer with persistent memory, session management, and MFA authentication. - Pioneered a hybrid FAISS + BM25 retrieval service with sentence-transformer embeddings and a Cohere/BGE reranker, reducing LLM hallucinations by 40-60% across 5 projects. - Built an LLM-powered Credit Card KYC Document Processing Pipeline for 12 distinct document types (Aadhaar, PAN, ReKYC forms, consent letters) using Gemini 2.5 Flash. - Benchmarked 7 face-embedding models (ArcFace, FaceNet variants on InsightFace) and shipped sub-100ms recognition at 99.2% accuracy under production load. - Fine-tuned Indic Parler-TTS and Wav2Vec2/Whisper STT and built ECAPA-TDNN speaker verification reaching 97% accuracy, plus anti-spoofing voice forensics.	
Tata Consultancy Services	Bengaluru, India
System Engineer	Mar 2022 – Aug 2023
- Developed a GPT-3-powered internal IT/HR helpdesk assistant, handling 5K+ daily employee queries (VPN, access requests, leave/payroll) with 4.3/5 user satisfaction. - Contributed to team-wide LLM safety checks - prompt- and regex-based input/output validation, PII redaction, and jailbreak detection for sensitive-data apps. - Engineered LLM observability dashboards tracking token usage, latency, cost, and error rates, enabling data-driven model selection cutting costs via open-source models for simple tasks.	
Ericsson	Bengaluru, India
Associate Engineer	Aug 2019 – Feb 2022
- Designed a hybrid recommendation system (collaborative filtering + DNN re-ranking) for device and accessory bundling on product pages, driving 23% CTR uplift across 5M users. - Adapted ResNet50/Inception via transfer learning to auto-categorize device catalog images across 10 categories, achieving high-90s% accuracy on frequent classes. - Fine-tuned BERT for ticket classification across 20 categories, auto-routing of high-frequency tickets with confidence-based fallback to manual queue, reducing SLA-impacting delays.	
DXC Technology	Bengaluru, India
Technical Solutions Rep	Mar 2018 – June 2019
- Trained and tuned Random Forest/XGBoost on ticket and shift-level features to predict escalation risk at intake, improving F1-score 12% over logistic regression baseline. - Engineered 40+ reusable features (staffing, ticket volume, resolution time, escalations) from raw SQL via Pandas/NumPy aggregations, cutting reporting/model prep time by 30%. - Rebuilt internal knowledge base into an SEO-indexed portal for a 40-person support team; improved search accuracy 20% and cut resolution time ~40% across 500+ monthly tickets.	

SKILLS

- **Languages** - Python, SQL, C++
- **GenAI / LLM** - LangChain, LangGraph, RAG, Agentic AI, MCP, LiteLLM, Prompt Engineering
- **ML / Deep Learning** - PyTorch, TensorFlow, Transformers, BERT, XGBoost, Pydantic
- **CV / Speech** - ArcFace, FaceNet, Whisper, Wav2Vec2, ECAPA-TDNN
- **Retrieval / Vector** - FAISS, BM25, Sentence-Transformers, pgvector, Hybrid Search, Reranking
- **Backend** - FastAPI, Flask, Node.js, REST APIs, Microservices
- **Databases** - PostgreSQL, YugabyteDB, Aerospike, Redis
- **Infra / Cloud** - Docker, Kubernetes, GCP (Pub/Sub), CI/CD (Jenkins), OpenTelemetry
- **Security / Auth** - MFA, PII Masking, OWASP LLM Top 10, Idempotency & IDOR Guards

EDUCATION

Georgian College	Barrie, Canada
Post Graduation, Major in Artificial Intelligence	Sept 2023 – Apr 2024
The Oxford College of Engineering	Bengaluru, India
Bachelor of Engineering, Major in Information Science	Aug 2013 – June 2017

HACKATHONS

MonsoonSaathi	July 2026
• Built a GenAI monsoon preparedness assistant - personalized plans, real-time alerts, multilingual support - ranking 16/128 at Google PromptWars Bengaluru (95.89/100 score).	
MediVision	Feb 2024
• Fine-tuned LLaVA-Med, an open-source vision-language model, to analyze injury images and suggest first-aid/OTC guidance, finishing 4th of 25 teams in a 24-hour AI hackathon.	

PROJECTS

AgentCLI	GitHub
• Built a CLI coding agent in Python supporting any OpenAI-compatible LLM endpoint with streaming tool-calling loop, multimodal input, context compaction, and session persistence.	
LabelMint	GitHub
• Built a self-hosted, open-source ML annotation tool - bbox/polygon labeling, redaction (blur/blacken/pixelate), review approve workflow, one-click export to YOLO/COCO/Pascal VOC.	

UPSKILLING

Bengaluru, India	Mar 2025 – Jun 2025
• Ran a focused job search for Senior AI/ML Engineer roles, sharpening System Design and DSA fundamentals while staying current on the evolving AI/ML stack.	
Toronto, Canada	May 2024 – Feb 2025
• Pursued Applied Scientist opportunities while deepening expertise in RAG and Agentic systems, reading and reproducing key ML research papers alongside part-time work.	